

DEXTER FERNANDES

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SUMMARY

Computer vision engineer with 3+ years shipping production ML systems end to end, from data curation and PyTorch detector training through quantisation, deployment and field validation on edge hardware. Led and mentored a team of four engineers, setting experimentation standards for CNN-based detection systems deployed on industrial safety and drilling-rig sites. Uses TIDE and confusion-matrix error analysis, ablation studies and benchmarking harnesses to close model performance gaps under real-time production constraints. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer 01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Benchmarked YOLOv5 against EfficientDet on a repeatable harness, quantifying accuracy, latency and stability pre and post optimisation to catch regressions before release.
- Ran TIDE error analysis on COCO-format production data to split live detector failures by error type, directing dataset fixes and threshold tuning at the dominant one.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.

LivNSense Technologies — Lead Software Engineer, AI 05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Set experimentation standards, reviewed model performance and mentored junior engineers, guiding optimisation strategy for production deployments.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Used confusion-matrix failure clustering and TIDE error analysis on COCO-format production data to diagnose recurring detector failures and prioritise dataset fixes across camera feeds.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review, foreground separation and failure analysis on industrial camera feeds.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline 06/2026

- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.
- Ran an O(1)-inference confidence-threshold sweep against WildTrack ground truth, lifting operating-point F1 from 0.218 to 0.591 and logging results to Weights & Biases.
- Built an automated model-promotion gate checking match-rate and mean-IoU against a signed SHA-256 manifest, and wired GPU smoke plus MOT17 integration tests into GitHub Actions CI.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking 03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-robot operation.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer Vision & DL	object detection, image classification, semantic and instance segmentation, object tracking, anomaly detection, defect detection, data augmentation, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, EfficientViT, Vision Transformers
Frameworks & Libraries	PyTorch, TensorFlow, Keras, OpenCV, scikit-image, NumPy, torchvision, Pillow, HuggingFace
Edge Inference	TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, VRAM profiling, Intel VTune, NVIDIA Jetson AGX Xavier, Intel Movidius VPU
Model Evaluation	precision and recall, confusion-matrix analysis, TIDE (Toolbox for Identifying Detection Errors), Grad-CAM, ablation studies, hypothesis testing, Bayesian optimisation, benchmarking harnesses, failure-mode analysis, drift tracking
MLOps	MLflow, DVC (Data Version Control), Apache Airflow, Apache Kafka, Optuna, Weights & Biases, Jupyter, CVAT (Computer Vision Annotation Tool), automated labelling, dataset versioning, synthetic data generation, model-promotion gates
Production & CI/CD	Docker, CI/CD, GitHub Actions, Git, CMake, pytest, unittest, gtest, REST API design, containerised microservices, multiprocessing, structured logging, health monitoring, observability, Linux
Cloud & Data	Microsoft Azure (Container Apps, Blob Storage, queues), PostgreSQL, autoscaling, cost-per-inference, Django
Classical ML	gradient boosting, random forests, scikit-learn, Pandas, multivariate time-series classification, MultiRocket
Professional	technical leadership, mentoring, Agile and Scrum, Jira, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering	05/2026 – PRESENT · REMOTE
Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.	

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)